A CLOSED SUBSCHEME OF A RELATIVE PROJ AWAY FROM A QUASI-COMPACT OPEN IS CUT OUT BY A FINITE TYPE GRADED PIECE

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 8144-8162 and proof span 8164-8187 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof refers to the auxiliary local data introduced in the proof of the earlier companion statement and to an integer d fixed there; the author's notation is preserved and no letter was renamed.

Editorial note. The proof contains no omissions or pointer-only steps; it is delivered as written.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The description of an arbitrary closed subscheme of a relative Proj as a relative Proj of the graded quotient by an explicitly constructed kernel, cited from the same chapter and established by a previous card of this lane. Cited at proof lines 8164-8166, together with the auxiliary data and the integer d produced in the proof of its finite type companion. Supplies the graded ideal whose d-th piece has to be replaced by a finite type submodule with the same effect.
- A quasi-coherent module over a quasi-compact quasi-separated scheme is the filtered union of its finite type quasi-coherent submodules, cited from the properties chapter. Cited at proof lines 8179-8181. Produces a single finite type submodule containing all the finitely many sections that have to be controlled.
- A finite type module over a quasi-compact scheme is generated by finitely many sections over each member of a finite affine cover. Used at proof lines 8174-8178. Supplies the sections g over the members of a finite affine cover of the given open that have to lie in the submodule.

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Source reference. Original lemma 0801: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10166–10166. Same-chapter reference used in the retained author text at source line(s) 7991, 8111, 8166. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0802: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10167–10167. Same-chapter reference used in the retained author text at source line(s) 8140, 8169. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01PG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2229–2229. Same-chapter reference used in the retained author text at source line(s) 8130, 8178. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let $\mathcal{A}$ be a quasi-coherent graded $\mathcal{O}_S$-algebra. Let $p : X = \underline{Proj}_S(\mathcal{A}) \to S$ be the relative Proj of $\mathcal{A}$. Let $i : Z \to X$ be a closed subscheme. Let $U \subset S$ be an open. Assume that*

1. *p is quasi-compact,*
2. *i of finite presentation,*
3. *$U \cap p(i(Z)) = \emptyset$,*
4. *U is quasi-compact,*
5. *$\mathcal{A}_n$ is a finite type $\mathcal{O}_S$-module for all n.*

Then there exists a $d > 0$ and a quasi-coherent finite type $\mathcal{O}_S$-submodule $\mathcal{F} \subset \mathcal{A}_d$ with (a) $Z = \underline{Proj}_S(\mathcal{A}/\mathcal{F}\mathcal{A})$ and (b) the support of $\mathcal{A}_d/\mathcal{F}$ is disjoint from U.

Proof. Let $\mathcal{I} \subset \mathcal{A}$ be the sheaf of quasi-coherent graded ideals constructed in Lemma 0801. Let U_i, R_i, A_i, I_i, $f_{i,j}$, $h_{i,j,k}$, and d be as constructed in the proof of Lemma 0802. Since $U \cap p(i(Z)) = \emptyset$ we see that $\mathcal{I}_d|_U = \mathcal{A}_d|_U$ (by our construction of $\mathcal{I}$ as a kernel). Since U is quasi-compact we can choose a finite affine open covering $U = W_1 \cup \ldots \cup W_m$. Since $\mathcal{A}_d$ is of finite type we can find finitely many sections $g_{t,s} \in \mathcal{A}_d(W_t)$ which generate $\mathcal{A}_d|_{W_t} = \mathcal{I}_d|_{W_t}$ as an $\mathcal{O}_{W_t}$-module. To finish the proof, note that by Properties, Lemma 01PG there exists a finite type $\mathcal{F} \subset \mathcal{I}_d$ such that all the local sections

$$h_{i,j,k} f_{i,j}^{(d-\deg(h_{i,j,k}))/\deg(f_{i,j})} \quad \text{and} \quad g_{t,s}$$

are sections of $\mathcal{F}$. By construction $\mathcal{F}$ is a solution. □